sAIs – Smart Autonomous Intelligent Parking System: Case Study Report

Conceptual engineering project: an intelligent autonomous parking and vehicle management ecosystem for a shopping mall

Status of this report. sAIs is a conceptual system proposed for a university engineering project. It has not been implemented or tested. Statements about benefits, performance and cost are expectations or planning assumptions, not measured results. PARKIE is an existing third-party robot; this report distinguishes its reported capabilities from the functions proposed for sAIs.

1. Introduction

Parking is the first and last service a shopping mall delivers to a visitor, yet it is still largely manual: drivers search for a free space, maneuver in constrained areas, and later try to remember where they parked. Intelligent autonomous parking replaces this manual process with a coordinated system in which robots move vehicles, sensors observe the facility, software allocates space, and the customer interacts through a mobile application [1], [3].

sAIs (Smart Autonomous Intelligent Parking System) is a proposed ecosystem that applies this concept to an indoor shopping mall. Customers leave their vehicle at a designated drop-off station, an autonomous parking robot transports it to a suitable space, and the customer later requests its return through a mobile application. Around the robot, sAIs adds an artificial intelligence layer, an Internet of Things (IoT) monitoring layer, a central management platform, an operator dashboard and customer services.

The purpose of this report is to present sAIs as a complete engineering solution: its problem context, architecture, components, operating workflow, intelligent features, integration approach, benefits, challenges and feasibility. Shopping malls are a suitable setting because they combine high and uneven demand, a strong link between customer experience and revenue, and a controlled indoor facility with a single operator [2], [4].

2. Background and Problem Statement

2.1 Limitations of traditional parking

In a conventional self-parking facility, every step depends on the driver. The literature on smart parking identifies recurring consequences of this model [1], [3], [4]:

- **Search time and congestion:** Drivers circulate looking for a space, which adds traffic inside the facility and delays entry and exit at peak times.

- **Inefficient space use:** Spaces and aisles are sized for human maneuvering and driver error, and unevenly parked vehicles leave unusable gaps.
- **Customer inconvenience:** Drivers walk from distant spaces, struggle in tight bays, and forget where they parked.
- **Limited visibility for operators:** Without real-time occupancy and vehicle data, operators cannot manage demand, plan staffing or respond quickly to incidents.
- **Minor damage and conflicts:** Maneuvering in narrow spaces increases the chance of door dings and low-speed collisions.

2.2 Why an intelligent autonomous solution is needed

Automating the physical movement of vehicles addresses the driver-dependent steps, but robots alone do not solve the management problem. Without a decision layer, a robot fleet would still need instructions on which space to use, which task to do first and how to respond to demand. Smart parking research stresses that sensing, data processing, prediction and communication must work together as one architecture [1], [3], and that data reliability, security, privacy and interoperability are critical design factors [4]. sAIs is therefore defined as a complete system rather than a single robotic device.

3. Project Description

3.1 The sAIs concept

sAIs is a proposed intelligent autonomous parking and vehicle management ecosystem for a shopping mall. It combines six elements: autonomous robotic parking, artificial intelligence, IoT sensing and monitoring, a central management platform, a customer mobile application, and optional additional services.

3.2 Proposed solution

A customer drives to a designated drop-off station, leaves the vehicle according to the safety procedure, and confirms the handover in the application. The central platform checks the vehicle, selects a suitable parking position and dispatches a parking robot. When the customer is ready to leave, a retrieval request in the application creates a retrieval task, and the robot delivers the vehicle to a pickup station.

3.3 Objectives

No.	Objective	Description	Indicative measure (proposed, not measured)
1	Reduce parking effort	Remove the need to search for a space and park manually	Time from arrival to leaving the vehicle

2	Automate vehicle movement	Use an autonomous robot for parking and retrieval	Share of tasks completed without manual intervention
3	Improve space organization	Allocate spaces using vehicle size, availability and robot travel distance	Utilization of available spaces
4	Provide real-time information	Monitor occupancy, vehicle status and robot status	Accuracy of availability information
5	Improve customer experience	Reservation, tracking, retrieval and notifications in one application	Retrieval request-to-pickup time; customer satisfaction
6	Support management	Give operators one platform for monitoring and analytics	Number of incidents detected and resolved
7	Enable future services	Allow EV charging, car washing, VIP and pricing features	Services integrated without redesign

The indicative measures are proposed key performance indicators. Target values must be justified by site analysis, supplier data or clearly labeled design assumptions.

3.4 Planning assumptions

Item	Assumption	Status
Facility	Indoor shopping mall parking	Planning scenario
Capacity	Hypothetical 500 vehicles	Planning scenario, not a confirmed design
Number of robots	Not finalized	Requires capacity analysis

Drop-off and pickup stations	To be determined	Requires capacity analysis
Robotic platform	PARKIE, an existing third-party product	Subject to supplier confirmation
sAIs	Conceptual system	Not implemented or tested

4. System Architecture

4.1 Architectural overview

sAIs is organized in layers. Customers and operators interact through the mobile application and the dashboard. The central management platform holds the business logic, the AI and analytics module and the parking database. An IoT layer supplies observations from the facility, and a robot management system, here PARKIE's, executes physical vehicle movement. The platform sits in the middle: it receives requests and observations, decides, and issues tasks. This layering follows the structure commonly described for smart parking systems [3].

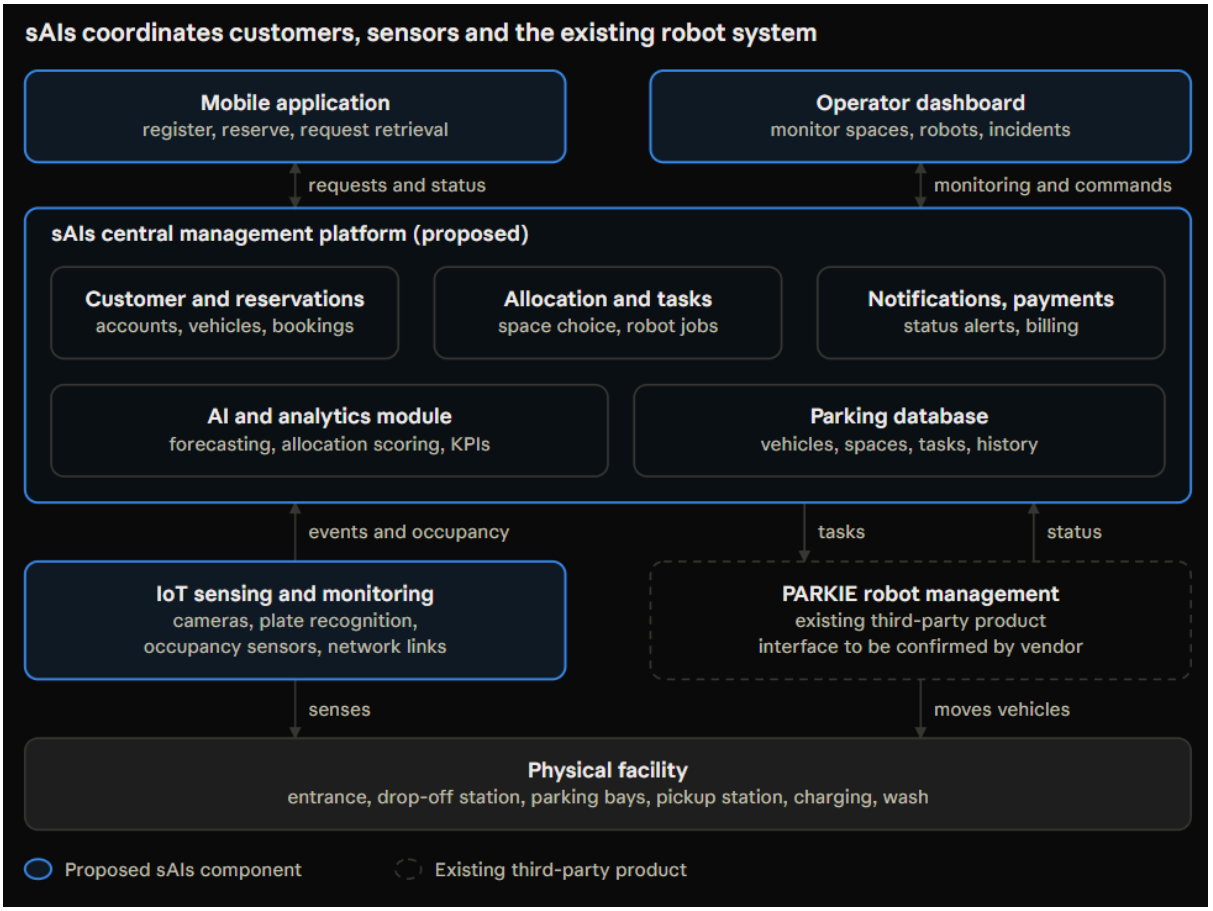


Figure 1. Proposed sAIs architecture. Proposed sAIs components are shaded; the existing PARKIE robot management system is dashed and its interface is subject to vendor confirmation.

4.2 Relationship between components

- **Mobile application:** The customer interface. It sends reservations, drop-off confirmations and retrieval requests, and receives status and location updates.
- **Central management platform:** The coordinator. It validates requests, allocates spaces, creates and tracks robot tasks, and triggers notifications.
- **AI and analytics module:** A decision-support service. It reads historical and real-time data from the database and returns forecasts, ranked space recommendations and performance analyses to the platform.
- **IoT sensors:** The observation layer. Cameras, license plate recognition and occupancy sensors send events to the platform, which updates the database.
- **Parking database:** The shared record of customers, vehicles, spaces, tasks, robot status and event history that all other components read and write through the platform.
- **Robotic parking system:** The execution layer. It receives parking and retrieval tasks, moves vehicles, and reports task progress and robot status.

4.3 Communication between components

The interfaces below are indicative design proposals. Final protocols depend on the technologies selected and on the interfaces the robot supplier confirms.

From	To	Data exchanged	Proposed mechanism
Mobile application	Platform	Reservations, handover confirmation, retrieval requests	Secure web API over HTTPS
Platform	Mobile application	Task status, vehicle location, notifications	Push notifications and API responses
IoT sensors	Platform	Plate reads, occupancy events, camera alerts	Event messages over the facility network
Platform	AI module	Historical and live data, allocation queries	Internal service interface

AI module	Platform	Forecasts, ranked space recommendations, analytics	Internal service interface
Platform	Robot management system	Parking and retrieval tasks	Vendor-supported API, to be confirmed
Robot management system	Platform	Task progress, robot status, faults	Vendor-supported API, to be confirmed
Platform	Dashboard	Occupancy, robot status, alarms, analytics	Real-time web interface

4.4 Architectural principles

- **Modularity:** The robot is reached through a single integration adapter, so the platform is not tied to one vendor's interface.
- **Separation of safety functions:** The robot's own navigation, obstacle detection and safety behavior remain with the robot system. sAIs may request a task or an emergency stop but does not replace those functions.
- **AI as an advisory layer:** AI outputs support allocation and planning. If the AI module is unavailable, the platform falls back to rule-based allocation.
- **Human oversight:** Operators can monitor, pause or override tasks from the dashboard.
- **Graceful degradation:** Manual parking and retrieval procedures remain available if the automated system is unavailable.

5. System Components and Technologies

The table separates what already exists from what sAIs proposes.

Component	Technology	Existing or proposed
Autonomous parking robots	PARKIE autonomous mobile robot and its management system	Existing, third party
Robot sensing and navigation	Onboard obstacle, space and path detection	Existing, part of the robot

Facility IoT monitoring	Cameras, plate recognition, occupancy sensors, networking	Proposed in sAIs
Central management platform	Allocation, task coordination, reservations, notifications	Proposed in sAIs
AI and analytics	Forecasting, allocation scoring, analytics	Proposed in sAIs
Mobile application	Customer registration, reservation, retrieval	Proposed in sAIs
Operator dashboard	Monitoring and control interface	Proposed in sAIs
Database and infrastructure	Parking database, server and cloud services	Proposed in sAIs
Security and communications	Access control, encryption, network design	Proposed in sAIs

5.1 Autonomous parking robots

sAIs uses PARKIE by HL Robotics as the selected robotic platform. According to the manufacturer, PARKIE is an indoor autonomous mobile robot that detects obstacles, open spaces and driving paths, identifies wheel spacing and the vehicle center, and lifts and transports the vehicle without additional guiding infrastructure [6]. Depending on the configuration, lifting may involve two robots working together; this must be confirmed with the supplier. Section 8 describes the integration in detail.

5.2 IoT sensors and monitoring

The IoT layer provides facility-level awareness that is separate from the sensors built into the robot. It includes cameras for monitoring and security, license plate recognition at the entrance and stations, occupancy sensing for spaces and stations, and a reliable network connecting these devices to the platform. Vehicle detection and communication technologies are central design choices in smart parking systems [3], [4]. Sensor selection and coverage would be determined by a site survey.

5.3 Artificial intelligence

The AI module analyzes historical occupancy and arrival data together with live events to support demand forecasting, space recommendation and analytics. Parking prediction is a well-studied area that covers availability, behavior and demand prediction with methods ranging from statistical models to deep learning [2]. Model selection for sAIs would depend on the data available and is left to the design stage.

5.4 Mobile application

The application supports customer registration, vehicle registration, reservations, drop-off confirmation, vehicle location and parking status, retrieval requests, notifications, parking history and payments.

5.5 Database and cloud or server infrastructure

The parking database stores customers, vehicles, reservations, parking spaces, tasks, robot status and event logs. Time-critical functions such as task dispatch benefit from a server located at the facility, while analytics and customer-facing services can run on cloud infrastructure. The hosting split is a design decision to be made with cost, latency and data-protection requirements in view.

5.6 Parking management dashboard

The dashboard shows available and occupied spaces, drop-off and pickup activity, robot availability and task status, faults and maintenance alerts, and demand trends, and lets authorized staff intervene in tasks.

5.7 Security and communication systems

Parking data includes vehicle identity and movement and is therefore sensitive; security, privacy and data reliability are recognized as critical design factors in smart parking [4]. The proposed measures are authentication and role-based access, encryption of data in transit and at rest, segmentation of the robot and IoT network from public-facing services, audit logging of vehicle handovers and task events, and a defined data retention policy that complies with applicable data-protection rules.

6. Operational Workflow

6.1 End-to-end process

Step	Actor	Action	System function
1	Customer	Arrives at the mall parking entrance	Entrance detection

2	sAIs	Identifies the vehicle by license plate recognition or another suitable method	IoT layer, platform lookup of customer and reservation
3	Customer	Proceeds to the designated drop-off station	Application and signage guidance
4	Customer	Leaves the vehicle per the safety procedure and confirms handover	Drop-off confirmation recorded
5	Platform	Checks vehicle details and available spaces	Vehicle record and compatibility check, occupancy lookup
6	AI module	Selects a suitable parking position	Allocation scoring by vehicle size, availability and robot travel distance
7	Robot system	Receives the task and transports the vehicle	Task dispatch through the integration adapter
8	Platform	Updates the parking database and confirms to the customer	Record update and notification
9	Customer	Requests retrieval in the mobile application	Retrieval request
10	Platform	Assigns the retrieval task	Task creation and robot selection
11	Robot system	Retrieves the vehicle and delivers it to the pickup station	Task execution and progress reporting
12	Customer	Receives a notification and collects the vehicle	Handover confirmation and record closure

The exact handover, safety checks and customer identification steps must be finalized with the robot supplier and the facility operator.

6.2 Customer journey

The customer sees four moments: arrival and recognition, handover at the drop-off station, a confirmation with vehicle status, and a retrieval request followed by a pickup notification. Between handover and pickup the customer can check the vehicle location and parking status in the application.

6.3 Vehicle identification and drop-off

The vehicle is identified at the entrance and checked against registered vehicles and reservations. At the drop-off station the customer positions the vehicle in the marked area, switches it off, secures it and leaves it according to the operating procedure. The handover is confirmed in the application and recorded. Before a task is created, the platform checks that the vehicle meets the compatibility limits agreed for the robot system. A vehicle that fails this check is redirected to manual parking.

6.4 Parking allocation

The platform requests a recommendation from the AI module, which ranks candidate spaces using vehicle size, space availability and robot travel distance. The platform confirms the choice against current occupancy, reserves the space in the database, and creates the parking task.

6.5 Robotic vehicle transportation

The task is sent to the robot management system. The robot navigates to the vehicle, positions itself, lifts the vehicle and transports it to the assigned space, then releases it and becomes available for another task. These physical operations are performed by the robot system; sAIs tracks progress and records the outcome.

6.6 Retrieval and pickup

On a retrieval request the platform creates a retrieval task, selects an available robot through the robot management system, and monitors the task. When the vehicle reaches the pickup station, the customer is notified and the space is released in the database.

6.7 Operator monitoring and exceptions

The operator dashboard shows occupancy, active tasks, robot status and alarms. If a task fails, a robot reports a fault, or a sensor detects an unexpected condition, the platform raises an alert. The operator can pause or reassign tasks, call the supplier or maintenance staff, and switch to manual procedures when necessary. Detailed fault handling is covered in the project risk management plan.

7. AI and Intelligent Features

The AI module is an additional decision-support layer. It does not replace the robot's navigation and safety functions.

Feature	Purpose	Inputs	Output
Parking demand forecasting	Estimate near-term and daily demand	Historical occupancy, time, weekday, holidays, live arrivals	Demand forecast for operators and for other features
Intelligent space allocation	Recommend a suitable space	Vehicle size, space availability, robot travel distance, constraints	Ranked list of spaces
Parking optimization	Keep the facility organized and robot movement efficient	Current layout, pending tasks, forecast	Allocation and task ordering suggestions
Arrival-time recommendations	Reduce peaks	Demand forecast, reservations	Suggested arrival times in the application
Operational analytics	Evaluate performance	Task logs, robot status, occupancy	Retrieval time, robot utilization, peak periods, faults
Dynamic pricing (optional)	Balance demand and revenue	Demand forecast, duration, zone	Price suggestions

Parking prediction literature distinguishes availability, behavior and demand prediction and reviews methods from statistical models to graph neural networks [2]. The sAIs forecasting module corresponds to the demand prediction category. Because no operating data exists yet, the module would start with simple, explainable methods and be evaluated against real data once the facility operates.

Dynamic pricing is optional. It must be checked against local rules, operator policy and customer acceptance before use, and prices should always be shown to the customer before confirmation.

8. Integration of PARKIE into sAIs

8.1 Why PARKIE was selected

PARKIE is the robotic platform and technical reference for sAIs because it matches the central requirement of the concept: autonomous indoor movement of vehicles without driver involvement. The manufacturer describes it as an indoor autonomous mobile robot that

needs no additional guiding infrastructure and supports API integration with parking and building management systems [6]. It has been reported in commercial and pilot use in Korea, including a robot valet service in an underground parking facility that uses plate verification, kiosk registration and a designated robot valet zone [8], and pilot operation announced in 2023 [9]. Operation at Gimpo Airport [7], a 2026 agreement to deploy units in residential complexes [10] and a CES 2024 award [11] have also been reported. The selection is a planning choice; other robot platforms could be assessed through the same adapter.

8.2 Relevant capabilities (existing, as reported by the manufacturer)

- Autonomous navigation and detection of obstacles, available spaces and driving paths [6].
- Identification of wheel spacing and vehicle center to lift and transport a vehicle [6].
- A mobile application for drivers to follow vehicle location and reserve a pickup, and API integration with parking and building systems [6].
- Lifting that may involve two robots working together, depending on the configuration; to be confirmed with the supplier.

The manufacturer also claims up to 30% more parking spaces in some configurations [6]. sAIs does not adopt this figure. Any space utilization gain must be established by a site analysis.

8.3 Responsibilities: existing product versus proposed sAIs software

Function	PARKIE (existing, third party)	sAIs (proposed)
Navigation, obstacle detection, positioning, lifting, transport	Yes	No, sAIs does not replace these
Robot-level safety behavior	Yes	No, sAIs may request a stop but does not override
Robot fleet and site management software	Supplier system	Integrated through an adapter
Customer registration, reservations, payments	Not the focus	Yes
Vehicle records and parking database	Not the focus	Yes

Space allocation and AI features	Not the focus	Yes
Facility IoT monitoring	Onboard robot sensors only	Yes, facility-level monitoring
Operator dashboard across services	Supplier site tools	Yes, unified view
Additional services (EV charging, car wash, VIP)	No	Yes, as optional extensions

8.4 How sAIs communicates with the robot management system

sAIs would send parking and retrieval tasks to the robot management system and receive task progress, robot status and fault information. The platform would do this through a single integration adapter that translates sAIs task objects into whatever interface the supplier provides, so that vendor-specific details stay in one place. The manufacturer states that API integration is supported [6], but this report does not assume access to any specific interface.

8.5 Vendor confirmation required

The following must be confirmed with HL Robotics before the integration design is finalized:

Item	Question to the supplier
API and documentation	Which interfaces exist, what data and commands they cover, and how they are documented
Licensing and cost	Terms and cost for integration, robots and software, by written quotation
Vehicle compatibility	Supported vehicle sizes, weights and wheel geometries for the proposed configuration

Site requirements	Floor, ramp, clearance, lighting, connectivity and layout requirements
Safety and fault handling	Emergency stop, recovery procedure, remote assistance, and fault reporting
Fleet sizing	Task cycle times and recommended robot count for the expected demand
Support and maintenance	Service levels, spare parts, training and software updates
Regulatory status	Approvals and classification of robotic parking for the intended location

9. Additional Services

The services below are optional extensions. They are not required for a first version and would be added after the core workflow is validated.

- **VIP reservations:** Customers reserve designated premium spaces or services in advance, subject to availability. The platform treats them as a space category in allocation.
- **EV charging:** A customer requests charging when parking. The platform assigns a charging-capable space and shows charging status where the charging equipment supports integration. Charging zones also require specific fire and electrical safety review.
- **Automated car washing:** A dedicated automated washing station, separate from ordinary parking spaces, is used. The customer requests a wash in the application, and the platform coordinates the vehicle transfer to and from the station using staff or the robot system, subject to site and supplier confirmation. Washing equipment is not installed in parking spaces.
- **Security and vehicle monitoring:** Cameras, access control, vehicle records and incident alerts support facility security. The handover log and the task history also provide a record for resolving damage or identity disputes.

10. Expected Benefits

The benefits below are expected outcomes of the concept. None has been measured, and each must be validated through site analysis and pilot data.

Benefit	Mechanism in sAIs	Basis
Reduced customer effort and parking search time	The customer hands over the vehicle at the drop-off station and does not search for a space	Expected
Improved parking organization	Central allocation and a live record of every vehicle and space	Expected
Potentially better space utilization	Robot placement and allocation by vehicle size and availability	Potential; depends on site and robot, to be analyzed
Automated vehicle transportation	Robot performs parking and retrieval	Expected, subject to vehicle and site compatibility
Better parking management	Real-time occupancy, task and robot monitoring on the dashboard	Expected
Improved customer experience	Reservations, tracking, retrieval requests and notifications in one application	Expected; to be assessed by user feedback
Data-driven decision-making	Forecasts and analytics on demand, retrieval times and robot use	Expected; depends on data quality

11. Technical and Operational Challenges

Challenge	Description	Proposed response
Robot compatibility and limitations	Not all vehicles may be liftable; limits on size, weight and geometry apply	Agree compatibility criteria with the supplier and check at drop-off
Vehicle safety	Handover, lifting and transport can cause damage if misaligned or mishandled	Positioning checks, handover photos and records, clear liability terms
Facility suitability	Layout, floor, ramps, clearances and connectivity may not suit the robot	Formal site feasibility assessment
Software integration	The robot interface may be limited or unavailable	Early vendor documentation, a single adapter, manual fallback
Cybersecurity and privacy	Vehicle identity and movement data are sensitive; control systems can be attacked	Access control, encryption, network segmentation, data minimization and retention policy
Maintenance	Robots, sensors and software require upkeep and spare parts	Preventive maintenance plan and service agreement
Robot fleet capacity	Number of robots depends on arrival and retrieval demand, task time, distances, charging and maintenance	Throughput and peak-period analysis before sizing

Customer adoption	Drivers must trust handing over their vehicle	Clear instructions, a visible safety procedure and a standard parking option
Financial feasibility	Capital and operating costs are not yet known	Formal quotation, labeled planning assumptions, phased scope
Regulation	Robotic parking can be classified as a mechanical parking facility with separate requirements, as noted for Korea [8]	Check applicable rules for the intended location

12. Feasibility Analysis

This analysis is preliminary and qualitative. It identifies what looks achievable and what must still be established.

12.1 Technical feasibility

The core technologies already exist: an autonomous indoor parking robot that is reported to be in commercial and pilot use [6], [8], [9]; mature approaches to vehicle detection and plate recognition [3]; and established parking prediction methods [2]. The sAIs software layers are conventional systems-integration work. The main technical uncertainty is the robot interface and site compatibility, not the underlying technology. ISO 23374-1 defines a framework for automated valet parking systems, including management functions such as vehicle and facility compatibility checks, remote assistance and recovery, and operation stop commands [5]. It addresses automated driving of unoccupied vehicles, so its applicability to a robot that lifts vehicles must be confirmed, but it is a useful reference for the sAIs management functions.

12.2 Operational feasibility

An indoor mall with dedicated drop-off and pickup zones suits the workflow. Operation depends on trained staff, clear customer procedures, a manual fallback, defined handling of unsuitable vehicles, and a plan for faults and maintenance. Peak demand handling depends on fleet and station sizing, which has not been done.

12.3 Economic feasibility (conceptual)

No cost or revenue figures are claimed. The cost structure can be described by category:

Type	Categories
One-time	Robots and licensing, facility preparation and designated zones, sensors and networking, application and platform development, integration and commissioning, training
Recurring	Maintenance and spare parts, supplier support, cloud and server services, staff, energy, software updates, security and data protection

Possible value sources include higher customer satisfaction, lower valet labor, better turnover, and possibly better space use. Whether these outweigh the costs requires supplier quotations and a demand model.

12.4 Summary and further investigation

Dimension	Preliminary assessment	Main condition
Technical	Plausible	Confirmed robot interface and vehicle compatibility
Operational	Plausible	Site assessment, fleet sizing, staff procedures, fallback
Economic	Not yet determinable	Quotations and a demand and cost model

Before implementation, the following need further investigation:

1. Supplier confirmation of API access, licensing, vehicle limits, site requirements and safety procedures.
2. A site survey of the intended facility.
3. A throughput and fleet-size analysis based on arrival and retrieval rates.
4. A cost and revenue model based on quotations.
5. Applicable regulations, insurance and liability arrangements.
6. Data protection requirements and a security design.
7. A pilot to measure the indicative KPIs in Section 3.3.

13. Conclusion

sAIs is a proposed intelligent autonomous parking and vehicle management ecosystem for shopping malls. It combines an existing autonomous parking robot with a central management platform, facility IoT monitoring, an AI decision-support module, a customer mobile application, an operator dashboard and optional services. The robot performs the physical movement of vehicles; sAIs provides the coordination, information and customer experience around it.

The expected value of sAIs lies in reduced customer effort, better organized and monitored parking, and data-driven operation. These are expectations, not results, and depend on supplier confirmation, site compatibility, adequate fleet sizing, and sound safety, security and privacy design. The feasibility analysis suggests that the concept is technically and operationally plausible, while its economic case cannot yet be determined.

Future development could include a pilot to measure the proposed KPIs, support for additional robot platforms through the integration adapter, deeper analytics and pricing models once real data exists, and integration of EV charging, washing and VIP services after the core workflow is validated.

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